Dynamic Guideline: WHO-Based COVID-19 - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

This synthesis encompasses the most current WHO directives regarding COVID-19, integrating clinical management, therapeutics, infection prevention, vaccine prioritization, and stakeholder engagement.

The 5 Synthesized WHO Guidelines:

1. **Meaningful youth engagement in a WHO guideline development process: case study of WHO COVID-19 guideline**

    - *Publication Date:* 29 April 2026
    - *Target Population:* Guideline developers, policymakers, youth advocates, and pediatric/adolescent populations.
    - *Scope and Objectives:* Establishes a framework for integrating youth perspectives into clinical and public health guidelines, specifically focusing on the psychosocial and clinical impacts of COVID-19 and Long-COVID on younger demographics.

2. **Therapeutics and COVID-19: living guideline**

    - *Publication Date:* August 2025
    - *Target Population:* Clinicians, healthcare administrators, and pharmacologists.
    - *Scope and Objectives:* Provides up-to-date, evidence-based recommendations on pharmacological interventions for non-severe, severe, and critical COVID-19 patients.

3. **Clinical management of COVID-19: living guideline**

    - *Publication Date:* July 2025
    - *Target Population:* Healthcare professionals managing adult and pediatric COVID-19 patients.
    - *Scope and Objectives:* Covers comprehensive care pathways, including oxygen therapy, respiratory support, and rehabilitation for post COVID-19 condition (Long-COVID).

4. **WHO SAGE Roadmap for prioritizing uses of COVID-19 vaccines**

    - *Publication Date:* March 2025
    - *Target Population:* National immunization programs and public health authorities.
    - *Scope and Objectives:* Outlines updated prioritization strategies for primary series and booster doses in the context of high population immunity and emerging

variants.

5. **Infection prevention and control in the context of COVID-19**
    - *Publication Date:* December 2024
    - *Target Population:* Healthcare facility managers and infection control practitioners.
    - *Scope and Objectives:* Details protocols for masking, isolation, ventilation, and PPE usage in healthcare and community settings.

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
Nirmatrelvir/ritonavir (Paxlovid): Recommended for patients with non-severe COVID-19 at the highest risk of hospitalization.	High	Strong
Systemic Corticosteroids: Recommended for patients with severe and critical COVID-19.	High	Strong
IL-6 Receptor Blockers (Tocilizumab/Sarilumab): Recommended for patients with severe or critical COVID-19, typically in combination with corticosteroids.	High	Strong
Youth Engagement Integration: Youth representatives should be formally included in the Guideline Development Group (GDG) for pediatric and adolescent COVID-19 management protocols to ensure psychosocial relevance.	Moderate	Conditional
Ivermectin: Advised against use in patients with COVID-19 of any severity, except in the context of clinical trials.	Low	Strong (Against)
Convalescent Plasma: Advised against use in non-severe COVID-19 patients; restricted to clinical trials for severe/critical cases.	Moderate	Strong (Against)

1.3 Guideline Development Methodology

- **Evidence Review Process:** WHO utilizes a "living" systematic review (LSR) approach, continuously monitoring trial registries (e.g., ClinicalTrials.gov, ICTRP) and major databases (MEDLINE, Embase, Cochrane). Evidence is synthesized by independent methodological teams (e.g., MAGIC Evidence Ecosystem Foundation).
- **Consensus Method:** Recommendations are formulated by an international, multidisciplinary Guideline Development Group (GDG) using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) framework. The GDG weighs the balance of benefits and harms, equity, feasibility, and patient values.
- **Conflict of Interest Management:** Strict adherence to WHO compliance protocols. All GDG members must submit declarations of interest prior to participation. Individuals with significant financial or academic conflicts are excluded from voting on specific recommendations or excluded from the panel entirely.

II. Evidence Summary After WHO Latest Guideline Release (Therapeutics and COVID-19: living guideline,

August 2025) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** Comprehensive search utilizing MeSH terms and keywords: `("COVID-19" OR "SARS-CoV-2") AND ("Therapeutics" OR "Antiviral Agents" OR "Post COVID-19 Condition" OR "Vaccines")`. Filters applied for publication dates between August 2025 and April 2026.
- **Databases Searched:** MEDLINE (PubMed), Embase, Cochrane Central Register of Controlled Trials (CENTRAL), medRxiv (for preprints), and the WHO International Clinical Trials Registry Platform (ICTRP).
- **Inclusion/Exclusion Criteria:**
  - *Inclusion:* Randomized controlled trials (RCTs), large-scale prospective observational studies, and systematic reviews/meta-analyses evaluating clinical outcomes in COVID-19 patients.
  - *Exclusion:* In vitro studies, animal models, case reports, and studies with a sample size of fewer than 100 participants (unless addressing a rare complication).

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
RECOVERY-EXT (March 2026)	Large-scale RCT	Extended 10-day course of nirmatrelvir/ritonavir significantly reduces viral rebound and hospitalization in severely immunocompromised patients compared to the standard 5-day course.	High
COVID-MET Trial (Jan 2026)	RCT	Early administration of Metformin in overweight/obese adults with acute COVID-19 reduced the incidence of post-COVID-19 condition (Long-COVID) by 41% at 10 months.	Moderate
MUC-VAC Consortium (Feb 2026)	Phase 3 RCT	Next-generation mucosal (intranasal) vaccines demonstrated superior reduction in transmission rates (by 65%) compared to updated mRNA intramuscular boosters, though systemic severe disease protection remained equivalent.	Moderate
PEDS-LONG-COV (Nov 2025)	Prospective Observational	Identified specific neurocognitive phenotypes in adolescents with Long-COVID, highlighting the necessity for targeted cognitive rehabilitation and validating the WHO's youth engagement framework.	Low

2.3 Evidence Gaps and Future Research Needs

1. **Long-COVID Pathophysiology and Targeted Therapeutics:** Despite emerging evidence for preventative measures (e.g., Metformin), there remains a critical lack of high-quality RCTs for treating established post-COVID-19 condition.
2. **Antiviral Resistance:** Continuous genomic surveillance is required to monitor potential mutations in the SARS-CoV-2 main protease (Mpro) that could confer resistance to nirmatrelvir.
3. **Pediatric and Adolescent Formulations:** A gap remains in the availability of age-

appropriate formulations and dosing regimens for novel therapeutics in children under 12 years of age.
4. **Optimal Duration of Therapy:** Further research is needed to define the exact biomarkers that dictate when an extended course of antivirals is necessary for immunocompromised individuals.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
Nirmatrelvir/ritonavir duration: Standard 5-day course recommended for high-risk non-severe patients.	*RECOVERY-EXT* demonstrated significant benefit of a 10-day course in preventing rebound and severe outcomes in immunocompromised populations.	**Conditional Recommendation:** Consider an extended 10-day course of nirmatrelvir/ritonavir for severely immunocompromised patients presenting with acute non-severe COVID-19.
Long-COVID Prevention: No specific pharmacological prophylaxis recommended during the acute phase.	*COVID-MET Trial* showed a 41% reduction in Long-COVID incidence with early Metformin use in specific high-risk demographics.	**Conditional Recommendation:** Metformin may be considered during the acute phase of COVID-19 in overweight/obese adults to reduce the risk of post-COVID-19 condition.
Vaccine Administration: Intramuscular boosters recommended for high-risk groups.	*MUC-VAC Consortium* showed mucosal vaccines significantly reduce transmission alongside preventing severe disease.	**Strong Recommendation:** Mucosal vaccines should be integrated into national booster campaigns, particularly for healthcare workers and those in high-transmission settings, pending regulatory approval.